

Fine-Grained Analysis of Nonparametric Estimation for Pairwise Learning

Junyu Zhou^{ID}, Shuo Huang^{ID}, Han Feng^{ID}, Puyu Wang^{ID}, and Ding-Xuan Zhou^{ID}

Abstract—In this article, we are concerned with the generalization performance of nonparametric estimation for pairwise learning. Most of the existing work requires the hypothesis space to be convex or a VC-class, and the loss to be convex. However, these restrictive assumptions limit the applicability of the results in studying many popular methods, especially kernel methods and neural networks. We significantly relax these restrictive assumptions and establish a sharp oracle inequality of the empirical minimizer with a general hypothesis space for the Lipschitz continuous pairwise losses. As an example, we apply our general results to study pairwise least squares regression and derive an excess population risk bound that matches the minimax lower bound for the pointwise least squares regression. The key novelty lies in constructing a structured deep ReLU neural network to approximate the true predictor, and in designing a targeted hypothesis space composed of networks with this structure and controllable complexity. Experiments validate the effectiveness of the proposed method. This example demonstrates that the obtained general results indeed help us to explore the generalization performance on a variety of problems that cannot be handled by existing approaches.

Index Terms—Deep neural networks, generalization analysis, learning theory, nonparametric estimation, pairwise learning.

NOMENCLATURE

Abbreviation	Expansion
$\mathcal{X}/\mathcal{Y}$	Input/output space.
$\mathcal{H}$	Hypothesis space.
d	Dimension of the input space.
n	Training sample size.

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r	Smoothness index of the true predictor.
$\mathcal{E}(f)/\mathcal{E}_z(f)$	Population/empirical risk.
$\hat{f}_z$	$\arg \min_{f \in \mathcal{H}} \mathcal{E}_z(f)$.
$\hat{f}_{\mathcal{H}}$	$\arg \min_{f \in \mathcal{H}} \mathcal{E}(f)$.
f_{ρ}	True predictor.
$\mathcal{E}(\hat{f}_z) - \mathcal{E}(f_{\rho})$	Excess population risk.
$\ f\ _{L^{\infty}(\mathcal{X} \times \mathcal{X})}$	$\sup_{x, x' \in \mathcal{X}} f(x, x') $.
$\ f\ _{L^2_{\rho}}$	$(\int f ^2 d\rho_x)^{1/2}$.

I. INTRODUCTION

PAIRWISE learning is an important learning task that has attracted much attention in the modern machine-learning society. Many real-world tasks, such as ranking in search engines, recommendation systems, and metric learning, are inherently pairwise. Unlike pointwise learning which studies a target model based on a pointwise loss function, pairwise learning refers to learning tasks in which the loss function takes a pair of examples as the input. Specifically, the goal of the pairwise learning is to learn a predictor $f : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ based on a pair of example $((x, y), (x', y'))$, whose performance is measured by a pairwise loss $\ell : \mathbb{R} \times \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$. Here, $\mathcal{X} \subset \mathbb{R}^d$ is the bounded closed input space and $\mathcal{Y} \subset \mathbb{R}$ is the output space. Fig. 1 illustrates the difference between pointwise learning and pairwise learning tasks. Notable pairwise learning tasks include metric and similarity learning [1], [2], [3], AUC maximization [4], [5], [6], ranking [7], [8], [9], and a minimum error entropy principle [10]. For instance, metric and similarity learning aims to study a metric $d(\cdot, \cdot)$ that estimates the distance or the similarity between a pair of observers. The performance of d on a pair example $((x, y), (x', y'))$ is measured by the loss $\ell(\tau(y, y')d(x, x'))$ with $\tau(y, y') = 1$ if $y = y'$ and $\tau(y, y') = -1$ else. AUC maximization aims to rank positive instances above negative ones which involves a loss $\ell(w; (x, y), (x', y')) = (1 - w^T(x - x'))_+ \mathbf{I}_{[y=1 \wedge y'=-1]}$ with $x, x' \in \mathbb{R}^d$ and $y, y' \in \{\pm 1\}$. It has been shown that pairwise learning has great advantages over traditional pointwise learning in modeling the relative relationships between sample pairs in many problems [4], [11], [12], [13].

Parametric models usually refer to models whose architecture is fixed so that the number of parameters does not grow with the sample size n . In contrast, nonparametric models, especially those based on structured deep networks [14], [15], [16], allows the architecture and the number of parameters grow with n or with the desired accuracy ϵ . This is particularly valuable in pairwise settings, where the relation between input pairs and output pReferences or scores can be highly complex and nonlinear. Besides, nonparametric

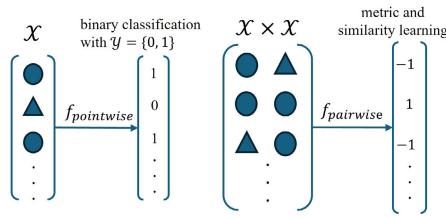


Fig. 1. Pointwise learning versus pairwise learning.

learning allows us to explore minimax-optimal learning rates with mild assumptions on the true data distribution or model structure. Despite the widespread use of pairwise learning, theoretical foundations, particularly the study of nonparametric models, remain underdeveloped. In this article, we focus on the generalization performance of nonparametric estimation for pairwise learning.

There has been a significant amount of work on the generalization analysis of nonparametric estimation for pairwise learning [9], [17], [18]. However, most of these works rely on restrictive assumptions regarding both the loss function and the hypothesis space, and typically focus on specific learning problems. For instance, Cl  men  on et al. [9] studied ranking problems and provided an oracle inequality for the empirical risk minimizer under the 0–1 loss, achieving a convergence rate of $O(((V \log(n))/n)^{1/(2-\beta)})$, where n is the training sample size, β is the parameter of the variance condition, and V is the VC-dimension of the class of the ranking rules. Here, the VC-dimension is the largest number of data points that can be labeled in all possible ways such that there exists a function in the class that correctly classifies each labeling, which measures the capacity or complexity of a class of functions. The requirement that the class of ranking rules has a finite VC-dimension is quite restrictive, as many hypothesis spaces—especially reproducing kernel Hilbert spaces (RKHSs)—do not satisfy this condition. Consequently, [9] are not applicable to widely used kernel-based methods. Furthermore, the 0–1 loss is rarely used in practice due to the associated minimization problem is known to be NP-hard even for a relatively simple class of functions such as linear function [19]. Rejchel [17] derived upper bounds for the estimation error in ranking problems under convex, nonnegative loss functions, assuming that the hypothesis space is convex. However, this convexity assumption limits the applicability of their results to neural networks, whose hypothesis spaces are typically nonconvex. Cao et al. [1] and Jin et al. [20] established estimation error bounds of order $O(n^{-1/2})$ for metric and similarity learning problems. More recently, [18] obtained excess risk bounds for metric and similarity learning with the hinge loss, exploiting the specific structure of the underlying true metric. Despite these advances, to the best of our knowledge, no existing work provides generalization analysis for pairwise learning with general loss functions under relaxed assumptions that accommodate modern hypothesis spaces, such as those used in neural networks or kernel methods.

On the other hand, numerous studies have applied neural networks to pairwise learning and ranking tasks [21], [22] [23], [24]. Nevertheless, a fundamental limitation of most existing models lies in their inability to capture nontransitive

pairwise interactions, a phenomenon observed in real-world scenarios but difficult to represent within the conventional score-difference framework. How to effectively model and learn from such nontransitive relationships remains an open and practically important question.

In this article, we investigate comprehensive generalization analysis of pairwise learning under general settings by significantly removing the restrictive assumptions on both the hypothesis space and the loss functions. Specifically, we establish an oracle inequality of the empirical minimizer for a *general hypothesis space* when the loss is Lipschitz continuous. Our results extend the existing literature in the following two aspects. First, the existing works either require the hypothesis space to be convex or a VC-class. As mentioned before, these assumptions are quite restrictive and cannot be applied to many popular methods including kernel methods and neural networks. Second, we consider the general losses that are Lipschitz continuous and possibly nonconvex. Various commonly used surrogate losses including the hinge loss, least squares loss, and logistic loss satisfy these assumptions, which makes our results applicable to a wide range of pairwise learning problems including ranking, pairwise regression, and metric and similarity learning. Furthermore, our experiments demonstrate that the proposed model is particularly well-suited for handling nontransitive pairwise relationships, providing a positive answer to the above mentioned question.

As an example, we apply our results to the study of pairwise least squares regression using deep ReLU networks. By leveraging our main theorem, we establish minimax-optimal excess population risk bounds of order $O(n^{-(2r)/(2r+d)})$ for pairwise least squares regression. Here, d is the dimension of the input space and $r \in \mathbb{N}$ is the smoothness index of the target function of the pointwise least squares regression. Our key idea is to construct a novel structured deep ReLU network (see Fig. 2) that approximates the true predictor f_ρ by exploiting the specific antisymmetric structure of f_ρ , and design the targeted hypothesis space that consisting of the networks with this structure. From a practical perspective, we provide a way to build such a structured deep ReLU network, which achieves superior generalization performance compared to standard benchmarks for the pairwise regression problem. Experimental results validate the effectiveness and advantages of our approach. This application shows that the general oracle inequality greatly helps us explore the generalization performance on a variety of problems that cannot be handled by existing approaches.

A. Organization of This Article

The rest of this article is organized as follows. Section II introduces some basic concepts and definitions. Section III presents the main result: an oracle inequality for the empirical minimizer over a general hypothesis space, under the assumption that the pairwise loss is Lipschitz continuous and considering various capacity conditions of the hypothesis space. In Section IV, we apply our main results to analyze the generalization performance of pairwise learning by constructing a structured deep ReLU network model. In particular, we derive minimax-optimal excess risk rates by leveraging the

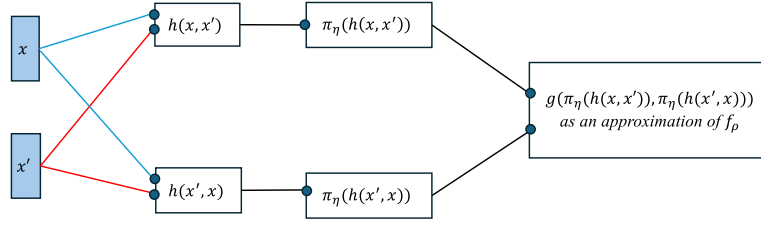
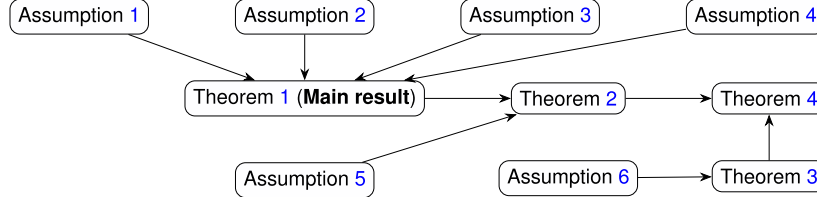

 Fig. 2. Structure of the designed antisymmetric deep ReLU network (8) with input $x, x' \in \mathcal{X}$.


Fig. 3. Diagram of the relationships among theorems and assumptions.

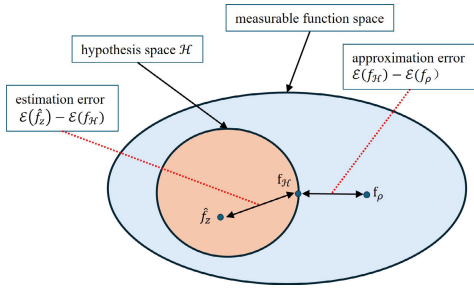


Fig. 4. Error decomposition.

main theorem. In Section V, we present experimental results to demonstrate the effectiveness of our structured deep ReLU model proposed in Section IV. We conclude this article in Section VI. To provide a clearer understanding of this article's theoretical structure, Fig. 3 illustrates the relationships among our theorems and assumptions. We give Nomenclature which summarizes the main notations.

II. LEARNING SETTING AND PRELIMINARIES

Let ρ be a population distribution defined on the sample space $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, where $\mathcal{X} \subset \mathbb{R}^d$ is the bounded closed input space and $\mathcal{Y} \subset \mathbb{R}$ is the output space. Let $\ell : \mathbb{R} \times \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ be a pairwise loss function. Given a predictor $f : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, the population risk $\mathcal{E}(f)$ with the loss ℓ is defined as

$$\begin{aligned} \mathcal{E}(f) &:= \int_{\mathcal{Z} \times \mathcal{Z}} \ell(f(x, x'), y, y') d\rho(z) d\rho(z') \\ &= \mathbb{E}[\ell(f(X, X'), Y, Y')]. \end{aligned}$$

Given a sample $S = \{Z_i = (X_i, Y_i)\}_{i=1}^n$ independently drawn from ρ , define the empirical risk based on S by $\mathcal{E}_z(f) := 1/(n(n-1)) \sum_{i \neq j} \ell(f(X_i, X_j), Y_i, Y_j)$. Let $f_\rho = \arg \min_{f \in \mathcal{F}} \mathcal{E}(f)$ be the true predictor (the target function) that minimizes the population risk over the space $\mathcal{F}$ consisting of all measurable functions from $\mathcal{X} \times \mathcal{X}$ to $\mathbb{R}$, and $\hat{f}_z = \arg \min_{f \in \mathcal{H}} \mathcal{E}_z(f)$ be the empirical minimizer with the hypothesis space $\mathcal{H}$. In this article, we are interested in studying the statistical generalization performance of $\hat{f}_z$, which is measured by the

excess population risk $\mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho)$, that is, the distance between the expected error of $\hat{f}_z$ and the least possible error $\mathcal{E}(f_\rho)$. We consider using the following error decomposition to study the excess population risk

$$\begin{aligned} \mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) &= \{\mathcal{E}(\hat{f}_z) - \mathcal{E}(f_{\mathcal{H}})\} + \{\mathcal{E}(f_{\mathcal{H}}) - \mathcal{E}(f_\rho)\} \\ &=: S(\mathcal{H}) + D(\mathcal{H}) \end{aligned} \quad (1)$$

where $f_{\mathcal{H}} = \arg \min_{f \in \mathcal{H}} \mathcal{E}(f)$. The terms $S(\mathcal{H})$ and $D(\mathcal{H})$ are called the estimation error and the approximation error, respectively, and will be estimated part by part. Fig. 4 illustrates this decomposition of the error. Throughout this article, we denote by $C_{\alpha, \beta, \gamma}$ the constant depends on parameters α, β , and γ , and denote C an absolute constant. These constants may differ from line to line and are always assumed to be greater than or equal to 1. We denote $\mathbf{1}_{[A]}$ the indicator function that takes value 1 if the event A happens and 0 otherwise. Let $\lceil C \rceil$ denote the least integer number greater than or equal to C . For a measurable set $A \subset \mathcal{X}$, let $\rho_X(A) := \text{Prob}\{A\} = \mathbb{E}_X[\mathbf{1}_{[A]}]$ and $\rho(A|x) := \text{Prob}\{A|X = x\} = \mathbb{E}[\mathbf{1}_{[A]}|X = x]$.

Our analysis requires some standard assumptions on the true predictor f_ρ , the loss function ℓ , and the capacity of the hypothesis space.

Assumption 1: There exists a constant $\eta > 0$ such that $\|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})} := \sup_{x, x' \in \mathcal{X}} |f_\rho(x, x')| \leq \eta$.

Most important problems satisfy this assumption. We use pairwise least squares regression and ranking problems as two examples here.

Example 1 (Pairwise Least Squares Regression): For the pairwise least squares regression with loss $\ell(f(x, x'), y, y') = (f(x, x') - y + y')^2$, it is known that the true predictor has the form $f_\rho(x, x') = \tilde{f}_\rho(x) - \tilde{f}_\rho(x')$, where $\tilde{f}_\rho(x) := \mathbb{E}[Y|X = x]$ is the target function (Bayes rule) for the pointwise least squares problem [25]. One often assumes that the distribution of Y is bounded by a constant $B > 0$, that is, $\text{Prob}\{|Y| \leq B\} = 1$, then we can show that Assumption 1 holds with $\eta = 2B$ by noting $\|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})} \leq 2\|\tilde{f}_\rho\|_{L^\infty(\mathcal{X})} \leq 2B$.

Example 2 (Ranking problems): For ranking problems with the hinge loss $\ell(f(x, x'), y, y') = (1 - \text{sgn}(y - y')f(x, x'))_+$, the work [26] proved that the true predictor has the form

$f_\rho(x, x') = \text{sgn}(T(x, x') - T(x', x))$, where $T(x, x') = \text{Prob}\{Y > Y' | X = x, X' = x'\}$. Then, it is easy to show that $\eta = 1$ in this case since $|\text{sgn}(t)| \leq 1$ for any $t \in \mathbb{R}$.

We also need to make the following assumptions about the Lipschitz continuity of the loss function. Here, we only consider the Lipschitz property of ℓ over $[-\|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})}, \|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})}]$ since the values predicted by f_ρ always lie on this interval.

Assumption 2: There exists a constant $K > 0$, such that for any $t_1, t_2 \in [-\|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})}, \|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})}]$ and $y, y' \in \mathcal{Y}$, there holds

$$|\ell(t_1, y, y') - \ell(t_2, y, y')| \leq K |t_1 - t_2|.$$

Assumption 2 holds for the hinge loss with $K = 1$ and the pairwise least squares loss with $K = 8B$ if the distribution of Y is bounded by $B > 0$.

With a little abuse of notation, for any probability measures ρ , we denote L_ρ^2 as the metric induced by the norm $\|\cdot\|_{L_\rho^2}$. Here, $\|\cdot\|_{L_\rho^2}$ is L^2 norm defined by $\|f\|_{L_\rho^2} = (\int |f|^2 d\rho_x)^{1/2}$. Recall that ρ_x is the marginal distribution of ρ on $\mathcal{X}$, we define two empirical probability measures based on the observed sample S as $\mu_n := 1/n \sum_{i=1}^n \delta_{X_i}$ and $\nu_n := 1/(n(n-1)) \sum_{i \neq j}^n \delta_{(X_i, X_j)}$, where $\delta_{(\cdot)}$ is the counting measure. For any $f \in \mathcal{H}$, we define the norms

$$\|f\|_{L_{\rho_x \times \mu_n}^2} := \left(\frac{1}{n} \sum_{i=1}^n \mathbb{E}[f(X, X_i) | X_i]^2 \right)^{\frac{1}{2}} \quad (2)$$

$$\|f\|_{L_{\nu_n}^2} := \left(\frac{1}{n(n-1)} \sum_{i \neq j} f(X_i, X_j)^2 \right)^{\frac{1}{2}}. \quad (3)$$

Our analysis needs the assumptions on the capacity of the hypothesis space $\mathcal{H}$, which are measured by different conditions of the associated covering numbers. The definition of covering number is given as follows.

Definition 1 (Covering Number [27]): Let $(\mathcal{T}, d)$ be a metric space. Consider a subset $\mathcal{K} \subset \mathcal{T}$ and let $\epsilon > 0$. A subset $\mathcal{N} \subset \mathcal{K}$ is called an ϵ -net of $\mathcal{K}$ if every point in $\mathcal{K}$ is within a distance ϵ of some point of $\mathcal{N}$, that is, $\forall x \in \mathcal{K}, \exists x_0 \in \mathcal{N} : d(x, x_0) \leq \epsilon$. The smallest possible cardinality of an ϵ -net of $\mathcal{K}$ is called the covering number of $\mathcal{K}$ and is denoted by $\mathcal{N}(\mathcal{K}, d, \epsilon)$.

Assumption 3: There exist constants $s_1, s_2 \geq e$ and $V_1, V_2 > 0$ such that for any $\epsilon \in (0, 1]$, there holds

$$\mathcal{N}(\mathcal{H}, L_{\rho_x \times \mu_n}^2, \epsilon) \leq s_1 \left(\frac{1}{\epsilon} \right)^{V_1}, \quad \mathcal{N}(\mathcal{H}, L_{\nu_n}^2, \epsilon) \leq s_2 \left(\frac{1}{\epsilon} \right)^{V_2}.$$

Any bounded subset of a finite-dimensional space [28] and the space consisting of bounded deep neural networks [14], [29] satisfy Assumption 3. However, many important spaces do not satisfy this assumption, for example, RKHSs. We introduce the following assumption as an alternative to Assumption 3 to handle this setting.

Assumption 4: There exist constants $s'_1, s'_2 \geq 1$ and $V'_1, V'_2 \in (0, 1)$ such that for any $\epsilon \in (0, 1)$,

$$\begin{aligned} \log(\mathcal{N}(\mathcal{H}, L_{\rho_x \times \mu_n}^2, \epsilon)) &\leq s'_1 \left(\frac{1}{\epsilon} \right)^{V'_1} \\ \log(\mathcal{N}(\mathcal{H}, L_{\nu_n}^2, \epsilon)) &\leq s'_2 \left(\frac{1}{\epsilon} \right)^{V'_2}. \end{aligned}$$

We further introduce the pseudo-dimension [29] and [30] of the capacity of the hypothesis space, which is defined by VC-dimension of a subgraph set. For a formal definition of VC-dimension, one can see [31, Definition 9.6].

Definition 2 (Pseudo-Dimension [29], [30]): Let $\mathcal{F}$ be a class of functions $f : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ and $\mathcal{F}^+ := \{(x, x', t) : f(x, x') > t, f \in \mathcal{F}\}$ be its subgraph set, then the pseudo-dimension $\text{Pdim}(\mathcal{F})$ of $\mathcal{F}$ is defined as $\text{Pdim}(\mathcal{F}) := \text{VC}(\mathcal{F}^+)$, where $\text{VC}(\mathcal{F}^+)$ is the VC-dimension of $\mathcal{F}^+$. Further, if $\text{Pdim}(\mathcal{F}) < \infty$, then we call $\mathcal{F}$ a VC-class.

The following lemma reveals a relation between the covering number and pseudo-dimension [32, Theorem 2.6.7] and will be used frequently later.

Lemma 1 ([32, Theorem 2.6.7]): For a VC-class $\mathcal{F}$ of functions with uniform bound F , one has for any probability measure ρ

$$\mathcal{N}(\mathcal{F}, L_\rho^2, \epsilon F) \leq C \text{Pdim}(\mathcal{F}) (16e)^{\text{Pdim}(\mathcal{F})} \left(\frac{1}{\epsilon} \right)^{2(\text{Pdim}(\mathcal{F})-1)}$$

for an absolute constant $C > 0$ and $0 < \epsilon < 1$, where L_ρ^2 denotes the L^2 norm with respect to ρ_x .

Lemma 1 implies that any bounded VC-class satisfies Assumption 3. Specifically, for any hypothesis space $\mathcal{H}$ which is a VC-class and uniformly bounded by $\eta > 0$, $\mathcal{H}$ satisfies Assumption 3 with $s_1 = s_2 = C_{\eta, \text{Pdim}(\mathcal{H})}$ and $V_1 = V_2 = 2(\text{Pdim}(\mathcal{H}) - 1)$, respectively. Conversely, if we further assume that the inequalities therein hold with any probability measures, then Assumption 3 can be employed as an alternative definition of a VC-class.

III. MAIN RESULTS

In this section, we present our main result on an oracle inequality of $\hat{f}_\epsilon$. To obtain a fast convergence rate, we require the following variance condition (the second-order property) of a function class [33].

Definition 3 (Variance-expectation bound [33]): Let $M > 0$ and $\beta \in [0, 1]$. Let $\mathcal{F}$ be a function class consisting of functions $f : \mathcal{Z} \times \mathcal{Z} \rightarrow \mathbb{R}$ with nonnegative first moment, that is, $\mathbb{E}[f] \geq 0$ for any $f \in \mathcal{F}$. We say that $\mathcal{F}$ has a variance-expectation bound with parameter pair (β, M) , if for any $f \in \mathcal{F}$

$$\mathbb{E}[f^2] \leq M (\mathbb{E}[f])^\beta. \quad (4)$$

In statistical learning theory, we often set $\mathcal{F}$ to be the shifted hypothesis space $\{\ell(f) - \ell(f^*) : f \in \mathcal{H}\}$, where f^* is either the true predictor f_ρ or the oracle $f_{\mathcal{H}}$ (best predictor in $\mathcal{H}$). The inequality (4) holds with $\beta = 0$ for any uniformly bounded function class. There are numerous cases in which (4) also holds with $\beta \in (0, 1]$. For instance, (4) is satisfied for ranking [9] and for binary classification [34] if the distribution ρ over $\mathcal{Z}$ satisfies a low-noise condition. [35] showed that (4) holds with $\beta = \min\{1, 2/r\}$ if the hypothesis space $\mathcal{H}$ is convex and the modulus of convexity δ of the loss satisfies $\delta(\epsilon) \geq c\epsilon^r$. Further, we will prove later (4) is satisfied with $\beta = 1$ for pairwise least squares regression when the distribution ρ is bounded (see Lemma 10 in Appendix B of Supplementary Material for more details).

Note in the following theorem, we remove several restrictive assumptions commonly used in existing literature, such as the

requirement that the hypothesis space be convex [17], [35] or form a VC class [9]. Specifically, part (b) of the theorem is established for non VC-class, and the whole theorem holds for arbitrary nonconvex hypothesis spaces satisfying mild capacity conditions. The proof is given in Appendix A of Supplementary Material. We denote $A \vee B = \max\{A, B\}$.

Theorem 1: Suppose Assumptions 1 and 2 hold. Let $\mathcal{H}$ be an arbitrary hypothesis space with uniform bound $\eta > 0$ such that for any $f \in \mathcal{H} \cup \{f_\rho\}$ and almost $z, z' \in \mathcal{Z}$, there holds

$$\ell(f(x, x'), y, y') = \ell(f(x', x), y', y). \quad (5)$$

Let $M > 0$ and $\beta \in [0, 1]$, suppose that the shifted hypothesis space $\mathcal{F} := \{\ell(f(x, x'), y, y') - \ell(f_\rho(x, x'), y, y') : f \in \mathcal{H}\}$ has a variance-expectation bound (4) with parameter pair (β, M) . The following statements hold true.

- (a) If the capacity of $\mathcal{H}$ satisfies Assumption 3, then for any $\delta \in (0, 1/2)$, with probability at least $1 - \delta$

$$\begin{aligned} & \mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) \\ & \leq C_{\eta, K, M, \beta} \left(\frac{(V_1 \vee \log(s_1)) \log(n)}{n} \right)^{\frac{1}{2-\beta}} \log\left(\frac{4}{\delta}\right) \\ & \quad + \frac{C_{\eta, K} \log^2\left(\frac{\delta}{2}\right) \max\{V_1, V_2, \log(s_1), \log(s_2)\}}{n} \\ & \quad + (\beta + 2) (\mathcal{E}(f_{\mathcal{H}}) - \mathcal{E}(f_\rho)). \end{aligned}$$

- (b) If the capacity of $\mathcal{H}$ satisfies Assumption 4, then for any $\delta \in (0, 1/2)$, with probability at least $1 - \delta$

$$\begin{aligned} & \mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) \\ & \leq \left(C_{\eta, K, M, \beta} \max \left\{ \sqrt{s'_1} \left(\frac{1}{n} \right)^{\frac{2}{(2+V'_1)(2-\beta)}} \right. \right. \\ & \quad \left. \left. \left(\frac{\log(n)}{n} \right)^{\frac{1}{2-\beta}} \log\left(\frac{4}{\delta}\right) \right\} + \frac{C_{\eta, K} \max\{s'_1, s'_2\} \log^2\left(\frac{\delta}{2}\right)}{n(1 - \max\{V'_1, V'_2\})} \right) \\ & \quad + (\beta + 2) (\mathcal{E}(f_{\mathcal{H}}) - \mathcal{E}(f_\rho)). \end{aligned}$$

Theorem 1 indicates that controlling the excess population risk reduces to analyzing the capacity or complexity of the hypothesis space and its approximation capability. Moreover, for a fixed hypothesis space, the first two terms vanish as the sample size $n \rightarrow \infty$. In this regime, the approximation error becomes the dominant factor determining the overall excess population risk. We note Theorem 1 establishes an oracle inequality for a broad class of loss functions, including the hinge loss, least squares loss, and logistic loss. This level of generality makes our results applicable to a wide range of pairwise learning problems (such as ranking, pairwise regression, and metric or similarity learning) across various settings, including kernel methods and neural networks.

Remark 1: We remark that if the hypothesis space $\mathcal{H}$ is a VC-class. From Lemma 1, we know $\mathcal{H}$ satisfies Assumption 3. Therefore, part (a) of Theorem 1 still holds for any VC-class $\mathcal{H}$ with V_1, V_2 replaced by $2(\text{Pdim}(\mathcal{H}) - 1)$, and s_1, s_2 be some constants, where $\text{Pdim}(\mathcal{H})$ is the pseudo-dimension of $\mathcal{H}$.

Besides ranking and pairwise least squares regression, our results can also be applied to learning problems whose predictor is independent of the order of input sample pairs,

for example, the metric learning and the similarity learning problems. Let us show how to apply Theorem 1 to metric learning problems. In distance metric learning, we aim to learn a Mahalanobis distance $f(x, x') = (x - x')^\top M(x - x')$, where $M \in \mathbb{S}_+^d$ is a positive semi-definite matrix. Let $\tau(y, y')$ be a function of labels such that $\tau(y, y') = 1$ if $y = y'$ and $\tau(y, y') = -1$ else. The performance of f on a sample pair (z, z') is measured by a loss function $\ell(\tau(y, y')(f(x, x') - b))$, where $b > 0$ is a bias term and ℓ is a convex and Lipschitz loss. The hypothesis space here is typically set to be $\mathcal{H} = \{(x - x')^\top M(x - x') : M \in \mathbb{S}_+^d, \|M\| \leq \eta\}$, where $\|\cdot\|$ denotes a regularization norm. Since the Mahalanobis distance and the function τ are both symmetric, then (5) holds. In addition, it is easy to verify that Assumptions 1 and 2 hold by noting that y is assumed to be bounded, and $\beta = 0$ due to the uniform boundedness of the hypothesis space. Note $\mathcal{H}$ is a bounded subset of a p -dimensional linear function space (with $p = d(d + 1)/2$). Then, Assumption 3 holds with $V_1 = V_2 = d(d + 1)/2$. Consequently, Theorem 1 yields an oracle inequality for metric learning in which the estimation error term scales as $O(1/\sqrt{n})$.

A. Comparison With Related Works

We give a comparison of our work with the existing results. The most related works are [9] and [17], which studied the generalization performance of ranking problems. Cl  men  on et al. [9] derived an oracle inequality of order $O(((V \log(n))/n)^{1/(2-\beta)})$ when the 0–1 loss is considered, where V is the VC-dimension of the class of the ranking rules. However, the 0–1 loss is not usually used in practice since the problem considered here is NP-hard and the empirical minimizer $\hat{f}_z$ is not easy to find through an efficient algorithm. Instead, we provide an oracle inequality with the same order (part (a) of Theorem 1) that holds for various commonly used surrogate losses including the hinge loss, least squares loss and exponential loss. This makes our results more broadly applicable. In addition, [9] required that the class of ranking rules is a VC-class, while many hypothesis spaces (e.g., RKHSs [36]) do not satisfy this assumption. We extend the desired result to a more general setting (part (b) of Theorem 1, where the hypothesis space is not assumed to be a VC-class), making our results tractable for a wider range of problems. Rejchel [17] established upper bounds for the estimation error with $\beta = 1$ under the assumption that the hypothesis space is convex. We investigate the excess population risk with parameter $\beta \in [0, 1]$. Moreover, the convexity assumption of the hypothesis space is not required here, which allows our results applicable to study the generalization performance of the neural networks where the hypothesis space is not convex in general. Cao et al. [1], Jin et al. [20], and Lei et al. [37] provided the estimation error bounds of order $O(n^{-1/2})$ for the metric and similarity learning problems. As mentioned before, Theorem 1 can also be applied to the metric and similarity learning. Very recently, several studies have further advanced the theoretical and algorithmic understanding of pairwise and metric learning [23], [38], [39], [40]. For example, [39] investigated pairwise learning with regularization networks and Nystr  m subsampling, providing algorithmic acceleration

TABLE I
COMPARISON OF DIFFERENT THEORETICAL SETTINGS IN RELATED WORKS

Work	Task	Loss	Capacity of $\mathcal{H}$	Convexity of $\mathcal{H}$
[1]	Metric and similarity learning	Hinge loss	VC-class	Convex
[17]	Ranking	Convex loss	VC or non VC-class	Convex
[26]	Ranking	Hinge loss	VC-class	Nonconvex
Ours	Metric and similarity learning, pairwise least squares regression ranking	General symmetric loss	VC or non VC-class	Convex or nonconvex

but still within shallow models. More recently, [23] established trajectory-dependent generalization bounds for pairwise learning with ϕ -mixing samples, extending generalization analysis to dependent data. Table I clearly summarizes and contrasts the theoretical settings of our work with those of previous studies, emphasizing the broader applicability of our framework.

IV. OPTIMAL RATES WITH DEEP RELU NETWORKS

In this section, we apply our main results to study the generalization performance of pairwise learning problems. Specifically, we show that pairwise least-squares regression can achieve the optimal excess population risk rate when the hypothesis class is chosen as a family of structured deep ReLU networks that matches the specific form of the true predictor.

A. Novel Approximation of the True Predictor

In this section, we first construct a novel structured deep ReLU network as an approximation of the true predictor f_ρ . Then, considering the hypothesis space consisting of the networks with this structure, we establish a sharp oracle inequality in the order of $O((\log(n))/n)^{1/(2-\beta)}$ for general antisymmetric losses. We will show in Section IV-B that, the excess risk rate that matches the minimax lower bound for least squares regression [14], [31] can be obtained when the least squares loss is considered.

In pairwise learning, we aim to learn a predictor $f : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ that models the relationship between two samples x, x' . In many applications such as pairwise ranking, the sign of $f(x, x')$ indicates whether x should be ranked ahead of x' . Since the prediction depends on the order of the inputs, it is natural to expect f to be anti-symmetric, that is, $f(x, x') = -f(x', x)$. The following assumption ensures that the true predictor f_ρ satisfies this antisymmetry property. The hinge loss $\ell(t, y, y') = (1 - \text{sgn}(y - y')t)_+$ for ranking and the least squares loss $\ell(t, y, y') = (t - y + y')^2$ for regression satisfy this assumption.

Assumption 5: The loss ℓ is anti-symmetric with respect to any predicted values and labels, that is, for any possible predicted value $t \in \mathbb{R}$ and labels $y, y' \in \mathcal{Y}$, there holds $\ell(t, y, y') = \ell(-t, y', y)$.

A good approximation of a model should be one that has the same structure as itself. Hence, the basic idea of designing an alternative of f_ρ is to construct a model that has the same structure as f_ρ . The following proposition shows that the true predictor has an antisymmetric structure under Assumption 5, which suggests the structure of the approximation of f_ρ .

The proof of Proposition 1 can be found in Appendix B of Supplementary Material.

Proposition 1: Under Assumption 5, the true predictor f_ρ is anti-symmetric, that is, for almost $x, x' \in \mathcal{X}$, there holds $f_\rho(x, x') = -f_\rho(x', x)$. Furthermore

$$f_\rho(x, x') = \frac{1}{2}f_\rho(x, x') - \frac{1}{2}f_\rho(x', x). \quad (6)$$

The nice decomposition of f_ρ given in the above proposition indeed tells us how to find a targeted approximation, that is, we consider making use of a series of ReLU networks to approximate f_ρ with the structure in (6). Before introducing our approximation, we give the definition of deep ReLU neural networks, which are the basis of our structured networks. We denote $\sigma(t) = (t)_+$ as the ReLU activation function acting componentwise on the vectors. Let $w_0 \in \mathbb{N}^+$ be the dimension of the input space, $w_L = 1$ and $w_l \in \mathbb{N}^+$ be the width of the l th layer for $l = 1, \dots, L-1$, where $L \in \mathbb{N}$ is the depth of the network. Let $\tilde{\mathcal{X}} \subset \mathbb{R}^{w_0}$. A deep ReLU neural network h from $\tilde{\mathcal{X}}$ to $\mathbb{R}$ with depth L has the form

$$h(\tilde{x}) = a^\top \sigma(T_{L-1} \sigma(T_{L-2} \cdots \sigma(T_1(\tilde{x})^\top + b_1) \cdots + b_{L-2}) + b_{L-1}) + b_L \text{ for any } \tilde{x} \in \tilde{\mathcal{X}} \quad (7)$$

where $T_l \in \mathbb{R}^{w_l \times w_{l-1}}$ indicates the connection matrix between the l th layer and the $(l-1)$ th layer, $b_l \in \mathbb{R}^{w_l}$ is the bias, and $a \in \mathbb{R}^{w_L}$. Let $\|\cdot\|_0$ denote the number of nonzero elements of the corresponding matrices and vectors. We define the number of nonzero weights, and computation units of h by $(\|a\|_0 + 1) + \sum_{l=1}^{L-1} \|T_l\|_0 + \|b_l\|_0$ and $\sum_{l=1}^{L-1} w_l$, respectively. We say a network h has the complexity (L, W, U) if its depth, the number of nonzero weights, and computation units are L, W , and U .

Now, we can give our structured approximation network $f : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ as follows:

$$f(x, x') = g(\pi_\eta(h(x, x')), \pi_\eta(h(x', x))) \quad (8)$$

for $x, x' \in \mathcal{X}$, where $\pi_\eta : \mathbb{R} \mapsto \mathbb{R}$ and $g : \mathbb{R} \mapsto \mathbb{R}$ are shallow ReLU networks with fixed complexity, and $h : \mathcal{X} \times \mathcal{X} \mapsto \mathbb{R}$ is a deep ReLU network defined in (7) with complexity (L, W, U) and $\tilde{\mathcal{X}} = \mathcal{X} \times \mathcal{X}$. The value of (L, W, U) will be properly chosen later for specific problems (see Theorem 4 for least squares regression as an example). Let us give some explanations of (8). First, we use a deep ReLU network h to approximate $1/2f_\rho(x, x')$ and $1/2f_\rho(x', x)$ by $h(x, x')$ and $h(x', x)$, respectively. Note that $\|f_\rho\|_{L^\infty(\mathcal{X} \times \mathcal{X})}$ is bounded by η , we hope that the values of h also lie in $[-\eta/2, \eta/2]$. Hence, we consider improving our estimate h by projecting the values

of h onto $[-\eta/2, \eta/2]$, that is, projecting h to $\pi_\eta(h)$ with the projection operator π_η defined as

$$\pi_\eta(t) := \begin{cases} \eta/2, & t > \eta/2 \\ t, & t \in [-\eta/2, \eta/2] \\ -\eta/2, & t < -\eta/2. \end{cases}$$

The operate $\pi_\eta(h)$ can be written as a shallow ReLU network: $\pi_\eta(t) = \sigma(t) - \sigma(t - \eta/2) - \sigma(-t) + \sigma(-t - \eta/2)$. Such an expression can be found in [41]. With this, two main items $1/2f_\rho(x, x')$ and $1/2f_\rho(x', x)$ appearing in (6) are addressed. It remains to find a ReLU network to handle the difference of the values of $\pi_\eta(h)$ between the sample pair (x, x') and its reverse order pair (x', x) . By noting that the difference operator can also be represented by a shallow ReLU network g , that is, $x - y = \sigma([1, -1] \cdot [x, y]) - \sigma([-1, 1] \cdot [x, y])$, we can give our final approximation $f(x, x') = g(\pi_\eta(h(x, x')), \pi_\eta(h(x', x)))$. Fig. 2 gives the specific structure of $f(x, x')$.

Note that the complexity of a network f of the form (8) can be computed by summing up the corresponding complexities of each subnetwork h, π_η , and g directly. Specifically, the depth of f is the summation of the depth of h, π_η , and g . The number of nonzero weights and computation units of f are $2(W_h + W_{\pi_\eta}) + W_g + c$ and $2(U_h + U_{\pi_\eta}) + U_g + c'$, respectively, where W_γ and U_γ denote the parameters of each subnetwork $\gamma \in \{h, \pi_\eta, g\}$, and $c, c' \in \mathbb{N}$ are absolute constants. The hypothesis space $\mathcal{H}$ consists of all possible predictors of the form (8) is defined as

$$\begin{aligned} \mathcal{H} &= \mathcal{H}(L, W, U) \\ &:= \{f \text{ of form (8): the complexity} \\ &\quad \text{of } f \text{ does not exceed } (L, W, U)\}. \end{aligned} \quad (9)$$

Here, we use parameters $(L, W, U) \in \mathbb{N}^3$ to measure the capacity (size) of $\mathcal{H}$. As these parameters increase, the capacity of the hypothesis is getting larger.

From [29, Th. 7], we know the hypothesis space $\mathcal{H}$ is a VC-class. Specifically, they proved that $\text{Pdim}(\mathcal{H}) \leq CLW \log(U)$. In the following theorem, we employ the pseudo-dimension to characterize the complexity of the hypothesis space. Unlike employing the L^∞ -covering number, the advantage of using pseudo-dimension is that any boundedness assumptions on the parameters of neural networks are not required. The proof of Theorem 2 can be directly derived by part (a) in Theorem 1. The detailed proof can be found in Appendix B of Supplementary Material.

Theorem 2: Suppose Assumptions 1, 2, and 5 hold. Let $\mathcal{H}$ be the space of structured deep ReLU networks (9) and $V = \text{Pdim}(\mathcal{H})$ be its pseudo-dimension. Let $M > 0$ and $\beta \in [0, 1]$, suppose that the shifted hypothesis space $\mathcal{F} := \{\ell(f(x, x'), y, y') - \ell(f_\rho(x, x'), y, y') : f \in \mathcal{H}\}$ has a variance-expectation bound with parameter pair (β, M) . Then for any $\delta \in (0, 1/2)$, with probability at least $1 - \delta$

$$\begin{aligned} \mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) &\leq C_{\eta, K, M, \beta} \left(\frac{V \log(n)}{n} \right)^{\frac{1}{2-\beta}} \log\left(\frac{4}{\delta}\right) \\ &\quad + C_{\eta, K} \frac{V}{n} \log^2\left(\frac{\delta}{2}\right) \\ &\quad + (\beta + 2) (\mathcal{E}(f_{\mathcal{H}}) - \mathcal{E}(f_\rho)). \end{aligned}$$

B. Pairwise Least Squares Regression With Deep ReLU Networks

In this section, we focus on the pairwise least squares regression with loss $\ell(f(x, x'), y, y') = (f(x, x') - y + y')^2$. We first estimate the approximation error with $\mathcal{H}$ of the form (9). Then, by combining the approximation error bound with Theorem 2 and choosing the proper capacity of $\mathcal{H}$, we derive the nearly optimal excess population risk bound for pairwise least squares regression.

The estimates of the approximation error $\mathcal{E}(f_{\mathcal{H}}) - \mathcal{E}(f_\rho)$ are closely related to the true predictor f_ρ , which is known to be $f_\rho(x, x') = \tilde{f}_\rho(x) - \tilde{f}_\rho(x')$ for $x, x' \in \mathcal{X}$ [25]. Here, $\tilde{f}_\rho(x) = \mathbb{E}[Y|X = x]$ is the regression function. To measure the smoothness of $\tilde{f}_\rho$, we need to introduce the Sobolev space. We first assume that $\mathcal{X} = [0, 1]^d$ in the remainder of this section. Let $r \in \mathbb{N}$ be the smoothness index, the Sobolev space $W^{r, \infty}([0, 1]^d)$ is defined as the class consisting of functions along with their partial derivatives up to order r lying in $L^\infty([0, 1]^d)$. The norm in $W^{r, \infty}([0, 1]^d)$ is defined as

$$\|f\|_{W^{r, \infty}([0, 1]^d)} := \max_{\alpha \in \mathbb{Z}_+^d: \|\alpha\|_1 \leq r} \|D^\alpha f\|_{L^\infty([0, 1]^d)}$$

where $\mathbb{Z}_+$ denotes the set of all nonnegative integer, $\|\alpha\|_1 = \sum_{i=1}^d |\alpha_i|$ denotes the l^1 norm of α , and $D^\alpha f = (\partial^{\|\alpha\|_1} f) / (\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d})$ denotes the weak partial derivative of f with order α .

Assumption 6 assumes all r th partial derivatives of the $\tilde{f}_\rho$ exist and their L^∞ norms are bounded.

Assumption 6: Suppose for some $r \in \mathbb{N}$, $\|\tilde{f}_\rho\|_{W^{r, \infty}([0, 1]^d)} \leq 1$. Since Sobolev norm dominates $L^\infty([0, 1]^d)$ norm, Assumption 6 implies Assumption 1 with $\eta = 2$.

The following theorem estimates the approximation error, whose proof is given in Appendix B of Supplementary Material.

Theorem 3: Suppose Assumption 6 holds, and the structured hypothesis space $\mathcal{H}(L, W, U)$ is defined by (9). Then, for any $\epsilon \in (0, 1)$, there exists a deep ReLU network f of form (8) with depth at most $C_{d, r} \log(1/\epsilon)$ and the number of nonzero weights and computation units at most $C_{d, r} \epsilon^{-d/r} \log(1/\epsilon)$ such that

$$\|f - f_\rho\|_{L^\infty([0, 1]^{2d})} \leq \epsilon.$$

Further, if we set $W = U = \lceil \exp(L) \rceil$, then the approximation error can be bounded as follows:

$$\mathcal{D}(\mathcal{H}) \leq C_{d, r} \left(\frac{L}{\exp(L)} \right)^{\frac{2r}{d}}.$$

Note that [42, Theorem 4, Part (b)] shows that, for a deep ReLU network f with depth at most $C_{d, r} \log(1/\epsilon)$, achieving an approximation error $\|f - f_\rho\|_{L^\infty} \leq \epsilon$ requires at least $C_{d, r} \epsilon^{-d/r} \log^{-3}(1/\epsilon)$ nonzero weights. Theorem 3 shows that a deep ReLU network with the same depth can achieve the same approximation error using only $C_{d, r} \epsilon^{-d/r} \log(1/\epsilon)$ nonzero weights. This implies that the bound in Theorem 3 is optimal up to the logarithmic factor $\log^4(1/\epsilon)$ in terms of the number of nonzero weights.

By combining Theorems 2 and 3, we can establish the following excess population risk bound of order $O(n^{-(2r)/(2r+d)})$,

which matches the minimax lower bound for the pointwise least squares regression [14], [31].

Theorem 4: Suppose Assumption 6 holds, $\text{Prob}\{|Y| \leq B\} = 1$ for some constant $B > 0$ and the structured hypothesis space $\mathcal{H}(L, W, U)$ is defined by (9). If we set $W = U = \lceil \exp(L) \rceil$, then for any $\delta \in (0, 1/2)$, with probability at least $1 - \delta$

$$\begin{aligned} \mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) &\leq C_{d,r,B} \frac{L^2 \exp(L) \log(n)}{n} \log^2\left(\frac{4}{\delta}\right) \\ &\quad + C_{d,r} \left(\frac{L}{\exp(L)}\right)^{\frac{2r}{d}}. \end{aligned}$$

Setting $L = \lceil d/(2r+d) \log(n) \rceil$, with probability at least $1 - \delta$, there holds

$$\mathcal{E}(\hat{f}_z) - \mathcal{E}(f_\rho) \leq C_{d,r,B} \log^\tau(n) n^{-\frac{2r}{2r+d}} \log^2\left(\frac{4}{\delta}\right)$$

where $\tau = \max\{3, 2r/d\}$.

1) *Comparison With Related Work:* The minimax lower rate $O(n^{-(2r)/(2r+s)})$ of the excess risk for pairwise learning is developed in [43] by using the kernel method, where $r \in (1/2, 3/2]$ is the smoothness parameter of f_ρ , and $s \in (0, 1)$ is the effective dimension measuring the capacity of the corresponding RKHS. Huang et al. [26] studied ranking with deep ReLU networks when the hinge loss is considered, they derived the excess risk rate $O(n^{-(r(\theta+1))/(2d+r(\theta+2))})$, where $r > 0$ denotes the smoothness of the target function, and $\theta > 0$ is the parameter of noise condition. As we mentioned before, the results of previous work [9], [17] cannot be used for studying the kernel methods or neural networks with the least squares loss, which demonstrates that our general results indeed help us to explore the generalization performance on a variety of problems that cannot be handled by existing approaches.

V. EXPERIMENTS

In this section, we present experimental results to demonstrate the effectiveness of our proposed model in both simulation and real-world data.

A. Evaluation of Approximation Error

This experiment aims to empirically verify the approximation error bounds stated in Theorem 3. The theorem provides upper bounds on the depth and number of nonzero weights required by deep ReLU networks to approximate smooth functions to a prescribed accuracy. To validate these results, we design a smooth target function and follow the basic idea of [42] and [44] to explicitly construct a deep ReLU network with controllable depth and sparsity to approximate it. By measuring the approximation error while increasing the network size (the depth and the number nonzero weights), we examine whether the observed convergence behavior matches the theoretical results.

1) *Target Function:* We consider a degree-7 B-spline interpolating function f_ρ defined on the interval $[0, 1]$. Note $f_\rho(x, x') = \tilde{f}_\rho(x) - \tilde{f}_\rho(x')$ in Theorem 3. In the following, we focus on approximating the target function in its pointwise form, that is, we consider $f_\rho(x)$. Specifically, 21 equally spaced

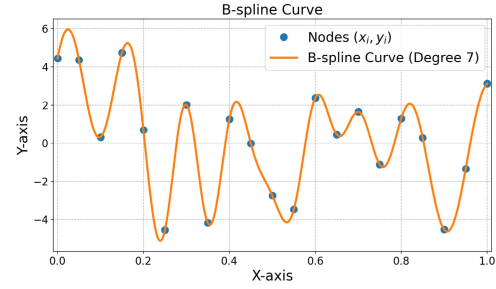


Fig. 5. B-spline of degree 7.

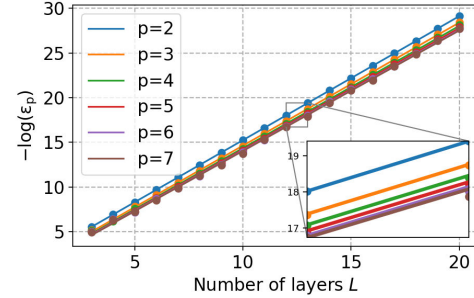


Fig. 6. Approximation accuracy $\epsilon_p = \|\psi_p(x) - x^p\|_\infty$.

nodes $\{x_i\}_{i=0}^{20}$ are chosen as $x_i = i/20$, and the corresponding function values $\{y_i\}_{i=0}^{20}$ are drawn independently of a uniform distribution on $[0, 10]$. The B-spline curve $f_\rho(x)$ is a smooth and piecewise polynomial, which can be uniquely determined by setting $f_\rho(x_i) = y_i$. That is, $f_\rho(x) = \sum_{p=0}^7 c_{i,p} x^p$ for any $x \in [x_{i-1}, x_i]$ and $i = 1, \dots, 20$. The target function f_ρ is illustrated in Fig. 5.

2) *Network Construction:* We aim to construct a deep neural network $\Psi(x)$ to approximate $f_\rho(x)$. To do so, we proceed in the following two steps. Step 1: approximate the power functions x^p by a network $\psi_p(x)$ for all $p = 0, \dots, 7$. Step 2: form linear combinations of $\psi_p(x)$ as $\sum_{p=0}^7 c_{i,p} \psi_p(x)$ approximate the target polynomial.

Step 1 (Approximating x^p). The basic idea to approximate x^p is drawn from [42] and [44]. First, we will find a L -layer deep ReLU network $\psi(x, x')$ that provides a good approximation of the product function $(x, x') \mapsto xx'$. Then, we can approximate x^2 by $\psi(x, x)$, that is, $\psi(x, x) \approx x^2$, and consequently x^3 can be approximated by $\psi(x, \psi(x, x))$. Following this idea, we can define a series of networks

$$\psi_1(x) = x, \quad \psi_p(x) = \psi(x, \psi_{p-1}(x)) \quad \text{for } p \geq 2$$

and use them to approximate x^p for all $p = 0, \dots, 7$.

Now, we explicitly construct the deep ReLU network $\psi(x, x')$ to approximate the product function $(x, x') \mapsto xx'$ on $[-1, 1]^2$. The network ψ with input $[x, x']^\top \in \mathbb{R}^2$ is defined layer by layer as follows: for the first and second hidden layers, we set their weight matrices as $T_1 \in \mathbb{R}^{4 \times 2}$, $T_2 \in \mathbb{R}^{6 \times 4}$ and bias $b_1 \in \mathbb{R}^4$, $b_2 \in \mathbb{R}^6$

$$T_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad b_1 = \mathbf{0}$$

TABLE II
 SUMMARY OF THE CONSTRUCTED NETWORKS

Approximator	Target	Depth	Nonzero weights
ψ	$(x, x') \mapsto xx'$	$O(L)$	$O(L)$
ψ_p	$x \mapsto x^p$	$O(pL)$	$O(pL)$
Φ	f_p	$O(pL)$	$O(p^2L)$

$$T_2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0 \\ -2^{-1} \\ 0 \\ 0 \\ -2^{-1} \\ 0 \end{bmatrix}.$$

For layers $\ell = 3, \dots, L+1$, each layer shares the same weight matrix $T_\ell = T^* \in \mathbb{R}^{6 \times 6}$

$$T^* = \begin{bmatrix} 0.5 & -1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.5 & -1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ -0.5 & 1.0 & 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.5 & -1.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.5 & -1.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & -0.5 & 1.0 & 1.0 \end{bmatrix}$$

and different bias $b_\ell = [0 \ -2^{-2\ell+3} \ 0 \ 0 \ -2^{-2\ell+3} \ 0]^T \in \mathbb{R}^6$.

The output layer has weight $T_{L+2} \in \mathbb{R}^{1 \times 6}$ and bias $b_{L+2} \in \mathbb{R}$

$$T_{L+2} = [-0.5 \ 1.0 \ 1.0 \ 0.5 \ -1.0 \ -1.0]$$

$$b_{L+2} = 0.$$

Let σ denote the ReLU activation function. The desired network $\psi(x, x')$ has the form

$$\psi(x, x') = T_{L+2} \sigma \left(T_{L+1} \sigma \left(\dots \sigma \left(T_1 [x, x']^T + b_1 \right) \dots \right) + b_{L+2} \right).$$

Step 2 (Linear Combination of x^p). For each x belongs to subinterval $[x_{i-1}, x_i]$, we can approximate $f_p(x)$ by $\Phi_i(x) = \sum_{p=0}^7 c_{i,p} \psi_p(x)$. Then, by concatenating all these subnetworks and denoting the resulting deep ReLU network by $\Phi(x)$, we obtain a global network that approximates f_p accurately over the entire domain. Table II summarizes the constructed deep ReLU networks.

3) Results: Fig. 6 shows how the approximation accuracy $\epsilon_p = \|\psi_p(x) - x^p\|_\infty$ varies as the number of layers L increases for each $p \in \{2, \dots, 7\}$. Here, ϵ_p denotes the infinity-norm between the subnetworks $\psi_p(x)$ and the power functions x^p . Since the infinity norm is taken over all $x \in [0, 1]$, we approximate it by taking the maximum absolute error over 20 000 uniformly spaced points on $[0, 1]$ [45]. In Fig. 6, the horizontal axis represents the number of layers L , and the vertical axis shows $-\log(\epsilon)$. A larger value of $-\log(\epsilon)$ indicates higher approximation accuracy. The plots reveal a striking linear relationship between $-\log(\epsilon)$ and L , indicating that the approximation error decreases *exponentially fast* with respect to the number of layers. Since the number of layers and nonzero parameters of ψ_p are of the same order (also see Table II), Fig. 6 implies that we only need $O(\log(1/\epsilon))$ number of layers and nonzero parameters to achieve the approximation accuracy ϵ . In addition, functions x^p with smaller degrees p exhibit better approximation accuracy.

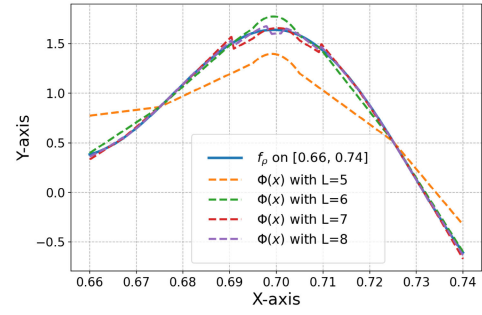
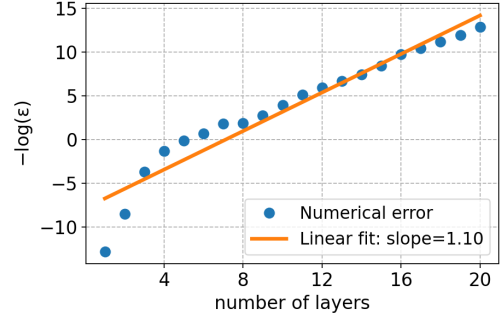
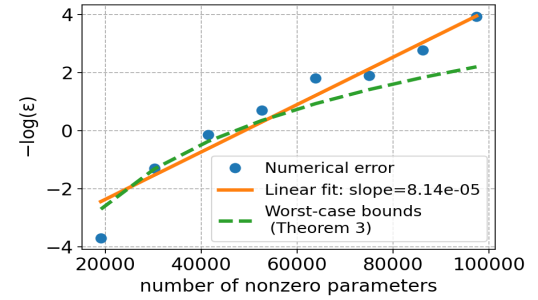


Fig. 7. Approximation behavior of the final deep ReLU network Φ with increasing layers.



(a)



(b)

Fig. 8. Scaling of approximation error with network size. Here, $\epsilon = \|\Phi(x) - f_p(x)\|_\infty$ and the numerical error (blue point) denotes the $-\log(\epsilon)$ as L and W increase. The orange line is the linear fit of the numerical errors, showing the approximation error decays exponentially as the number of parameters increases. (a) $-\log(\epsilon)$ versus the number of layers. (b) $-\log(\epsilon)$ versus the number of nonzero parameters.

Fig. 7 illustrates the local approximation behavior of the constructed deep ReLU networks Φ for different number of layers $L = 5, 6, 7, 8$. We focus on a representative subinterval $[0.66, 0.74]$. The blue solid curve corresponds to the B-spline f_p , while the dashed curves represent the network outputs $\Phi(x)$ with different depths. From Fig. 7, we can see that as the number of layers L increases, the approximation becomes progressively more accurate. Specifically, for $L = 5$, the network still exhibits visible deviation from the target, whereas for $L \geq 7$, the curves nearly coincide with f_p , capturing both the peak and the shape of f_p with high precision.

Fig. 8 quantitatively verifies the approximation error bounds in Theorem 3. Theorem 3 states that, to approximate a target function by deep ReLU network with accuracy ϵ , one needs to take the number of layers $L = O(\log(1/\epsilon))$ and

number of nonzero weights $W = O(\epsilon^{-c_0})$ at most, where $c_0 > 0$ is a constant. Fig. 8(a) shows an almost perfectly linear relation between $-\log(\epsilon)$ and L , matching the theoretical scaling $L \sim \log(1/\epsilon)$. Fig. 8(b) also exhibits a linear relation between $-\log(\epsilon)$ and W , implying $\epsilon \approx \exp(-\alpha W)$ (equivalently, $W \approx \alpha^{-1} \log(1/\epsilon)$), which is smaller than the worst-case requirement $W \sim \epsilon^{-c_0}$ (green dashed curve). The gap arises from the fact that our target function f_ρ is fixed, whereas the worst-case parameter requirements in Theorem 3 are established for approximating arbitrary f_ρ .

B. Nontransitive Pairwise Interactions

The aim of this experiment is to show the superiority of our model compared to the well-known RankNet. We model the pairwise preference probability between two items x, x' as

$$P_{x,x'} = P(y > y' | x, x') = \sigma(\beta [s(x) - s(x')] + \gamma R(x, x'))$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the logistic function. The first term $s(x) - s(x')$ corresponds to the standard score difference used in pairwise ranking models such as RankNet. The second term $R(x, x')$ is a pairwise, antisymmetric interaction term that cannot be represented as the difference of two univariate functions. The hyperparameters β and γ control the relative contributions of these two components. Unlike the score difference term, $R(x, x')$ can be nontransitive, that is, there exist items x_A, x_B, x_C such that

$$R(x_A, x_B) > 0, R(x_B, x_C) > 0, \text{ but } R(x_A, x_C) < 0.$$

As a result, when γ is relatively large compared to β , the model can produce cyclic preferences

$$P_{x_A, x_B} > \frac{1}{2}, \quad P_{x_B, x_C} > \frac{1}{2}, \quad P_{x_A, x_C} < \frac{1}{2}$$

corresponding to $x_A > x_B > x_C > x_A$.

1) *Preference Probability Setup*: We set the score function $s(x) = 2 \sin(2\pi x)$. Let $c_1 = 0.25$, $c_2 = 0.50$, $c_3 = 0.75$, and bandwidth $\tau = 0.06$. Define radial basis function (RBF) memberships and the antisymmetric matrix

$$\alpha_r(x) = \frac{\exp(-(x - c_r)^2 / (2\tau^2))}{\sum_{i=1}^3 \exp(-(x - c_i)^2 / (2\tau^2))}, \quad r \in \{1, 2, 3\}$$

and

$$K = \begin{pmatrix} 0 & +5 & -5 \\ -5 & 0 & +5 \\ +5 & -5 & 0 \end{pmatrix}.$$

Then, we define the antisymmetric interaction term $R(x, x') = \sum_{i,j=1}^3 \alpha_i(x) K_{ij} \alpha_j(x')$.

2) *Experiments Setup*: Inputs x are i.i.d. from a mixture on $[0, 1]$ (70% three Gaussians centered at (c_1, c_2, c_3) with std 0.06, 30% uniform). Pairs (x, x') are sampled uniformly. For each pair, we draw $U_1, \dots, U_{10} \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$, compute $\bar{U} = 1/10 \sum_{m=1}^{10} U_m$, and assign the label

$$y = \begin{cases} +1, & \bar{U} < P_{x,x'} \\ -1, & \text{otherwise.} \end{cases}$$

This yields low-variance stochastic labels. We generate 80 000 training pairs and 40 000 testing pairs from the above distribution. Our proposed model uses the pairwise function

$f(x, x') - f(x', x)$, and RankNet uses the score difference $g(x) - g(x')$. Both the underlying neural functions f and g are fully connected three-layer ReLU networks with a hidden width of 16. We train the models using the Adam optimizer with a batch size of 4000 and a learning rate of 4×10^{-3} . Each model is trained for 150 epochs, and all experiments are repeated five times, we report the averaged results over these runs.

3) *Results*: Fig. 9 shows the testing accuracy curves across different (β, γ) settings. Specifically, we vary $\beta \in \{0, 2\}$ and $\gamma \in \{2, 1, 0.5, 0\}$, where larger β emphasizes the univariate scoring difference component $s(x) - s(x')$, and larger γ emphasizes the pairwise interaction $R(x, x')$. Our results show that when $\beta = 0.0$ and $\gamma = 2.0$, RankNet achieves an accuracy close to 50%, which is equivalent to random guessing, indicating that it fails to learn any meaningful structure in the absence of score differences. However, our model fits the data well and achieves high accuracy. When $\beta = 2$ and γ is large, our model again significantly outperforms RankNet, which cannot represent the pairwise interaction structure induced by R . As γ decreases, the pairwise effect weakens and the performance gap narrows. When $\gamma = 0$, both models coincide and achieve similar accuracy as expected.

C. Real-World Data

In this experiment, we evaluate our model and the model proposed in [26] on the MQ2008 dataset (fold 1, LETOR). Since Huang's model is only defined under the hinge loss, we adopt the hinge loss for both models to ensure a fair comparison.

1) *Dataset and Protocol*: Each document is represented by 46 features. Pairs are constructed under the standard pairwise learning-to-rank protocol on the TRAIN split. We report *test accuracy* per epoch on the TEST split and average all curves over five independent runs (mean across seeds).

2) *Experiments Setup*: We compare our model $F(x, x') = f(x, x') - f(x', x)$ with $F(x, x') = h_\xi(f(x, x') - f(x', x))$ proposed in [26] using matched capacity. Here, $h_\xi(t) = 1/\xi(\sigma(t + \xi) - \sigma(t - \xi) - \xi)$ is the one-layer ReLU network used to approximate $\text{sign}(t)$ and $\xi = O((\log^2(n))/n^c)$, where n is the sample size and $c \in (0, 1)$ is some constant. Note that the parameter ξ in [26] decreases to 0 as the sample size increases. Then, we vary $\xi \in \{0.01, 0.005, 0.001\}$. Both models employ a two-hidden-layer neural function $f(x, x')$ with widths (256, 100) and ReLU activations, followed by dropout rates (0.5, 0.2). We optimize the *pairwise hinge loss* with margin $m = 1.0$ using Adam optimizer (learning rate 3×10^{-4}). Mini-batch size is set to a fixed fraction of the training set (batch_ratio = 1/20). Training runs for ten epochs, and results are averaged across five repeats.

3) *Results*: Fig. 10 shows the test accuracy versus training epochs for our model and [26] under different values of the outer ReLU parameter ξ . Our model consistently achieves the highest and most stable test accuracy (around 0.84) across all epochs. For model proposed in [26], the performance gradually degrades as ξ decreases. When $\xi = 0.01$, the model performs reasonably well, but as ξ further decreases to 0.005 and 0.001, both the average accuracy and the stability

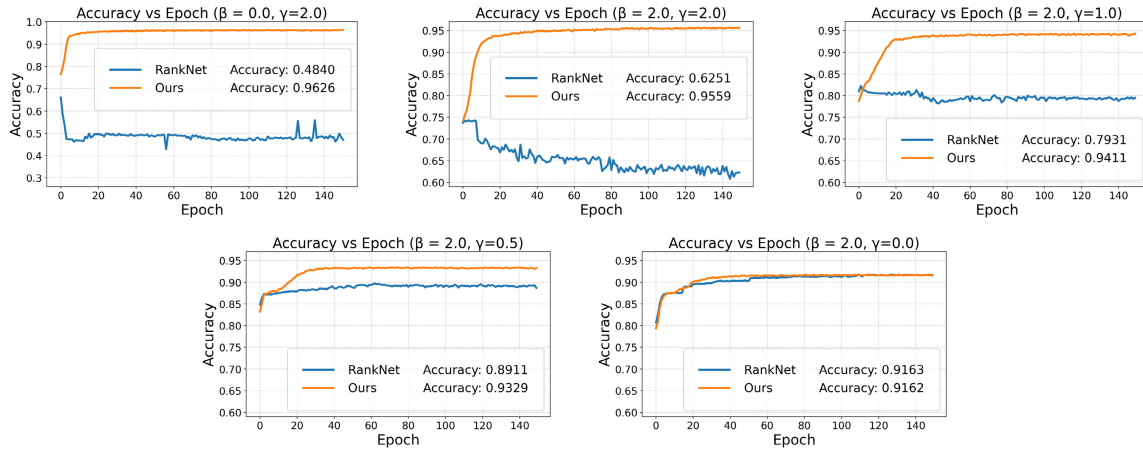


Fig. 9. Testing accuracy curves on synthetic data with different settings of β (score strength) and γ (pairwise interaction strength). RankNet fails when the pairwise interaction dominates ($\gamma = 2, \beta = 0$) or is strong ($\gamma = 2$ or $1, \beta = 2$), whereas our model consistently achieves high accuracy. When $\gamma = 0$, both models coincide.

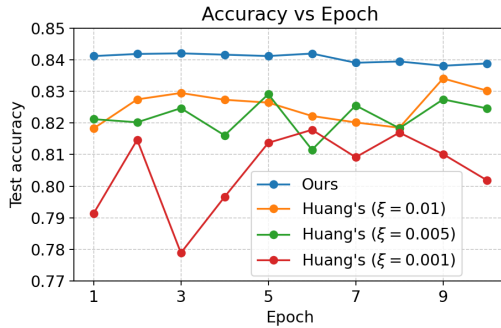


Fig. 10. Test accuracy versus epoch for our model and [26] with different ξ values.

across epochs noticeably decline. This trend aligns with the theoretical observation that the outermost ReLU in [26] is used to approximate the discontinuous $\text{sign}(x)$ function, where the approximation parameter $\xi = O((\log^2 n/n)^c)$ shrinks to 0 as the sample size increases. Consequently, a smaller ξ makes the activation more sharply nonlinear and difficult to optimize, leading to unstable training and inferior generalization. In contrast, our model avoids this sensitivity to ξ and maintains superior generalization performance, confirming the robustness and effectiveness of our design.

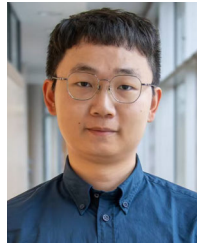
VI. CONCLUSION

In this article, we provide generalization analysis of pairwise learning under general settings. Specifically, we establish an oracle inequality of the empirical minimizer in the order of $O((\log(n)/n)^{1/(2-\beta)})$ for a general hypothesis space with the Lipschitz continuous and symmetric loss. Our general results greatly relax the conditions in previous works and can be applied to various learning problems. As an example, we apply our general results to conduct comprehensive generalization analysis for pairwise learning with structured deep ReLU networks. In particular, the excess population risk bound of order $O(n^{-(2r)/(2r+d)})$ that matches the minimax lower bound is achieved for pairwise least squares regression. Experiments show the effectiveness and efficiency of our proposed methods.

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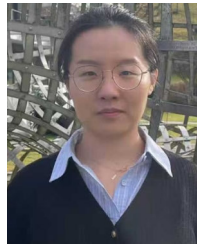
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